RSB Soil Impact Indicator Methodology: Soil carbon accumulation measurement and verification

Roundtable on Sustainable Biomaterials

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1. Introduction

It is widely known that increasing soil carbon stocks in agricultural fields is necessary to drive positive environmental benefits and mitigate the negative impacts of climate change.

The Roundtable on Sustainable Biomaterials (RSB) is a global membership organisation that drives the just and sustainable transition to a bio-based and circular economy. Our sustainability framework, underpinned by 12 key principles and their underlying criteria, is the world's most trusted and peer-reviewed foundation for supporting innovative solutions to the climate crisis. Application of our sustainability framework demonstrates a commitment to realising the United Nations (UN) Sustainable Development Goals. RSB's 12 Principles & Criteria (P&C[1]) describe the best social, legal, environmental and management practices for sustainable production in a bio-based and circular economy.

In response to global concerns over quality water quality and availability, RSB has defined Principle 9 within its P&Cs, directly addressing this matter. RSB Principle 9 "Water" requires RSB Participating Operators (POs) to maintain or enhance the quality and quantity of surface and groundwater resources, and respect prior formal or customary water rights.

While traditional certification focuses largely on risk management, an increasing number of buyers and corporations are setting quantifiable sustainability impact targets, seeking reliable claims beyond carbon mitigation within their supply chains. Demonstrating this shift, the Science-Based Targets Network has recently highlighted companies publicly adopting targets for freshwater and land, indicating a growing interest in measurable impacts beyond carbon mitigation.

Building on our established sustainability framework and rigorous multi-stakeholder consultation and validation process, RSB is well-positioned to support the creation of verifiable, positive impact claims to complement RSB certification. In 2023, RSB launched the RSB Impact Programme with the support of Boeing Global Engagement to bridge real positive sustainability impacts with market demand and facilitate the bio-based and circular economy transition. This programme meets the rising demand from various partners for 'in-sector' sustainability claims encompassing areas like soil health, water stewardship, and biodiversity. Key objectives include:

- Develop impactful claims: Use science-based methodologies to capture the impact of sustainable practices in metrics and translate them into powerful claims for companies and brands.
- Justify premiums for sustainability: Provide evidence of the value of sustainable feedstock and products in quantifiable metrics, encouraging private sector players to invest in sustainability.

[1] Roundtable on Sustainable Biomaterials: RSB Principles & Criteria (RSB-STD-01-001) (https://rsb.org/wp-content/uploads/2024/05/rsb-principles-criteria-std-01-001-v4-1.pdf)

- Create best-practice examples: Develop examples of motivated private sector involvement driving positive impacts throughout the bioeconomy.
- Impact Book & Claim: Enable brands to access impact claims related to their supply chain via the RSB Book & Claim System to support in-sector system transformations.

This document outlines the methodology for RSB's Soil Impact Indicator, focusing on soil carbon.

The approach: increasing soil carbon stocks through adopting conservation agricultural practices in sugarcane and grain crops cultivation

Scientific research has demonstrated the link between soil carbon stocks and adopting conservation agriculture practices in sugarcane and grain crop cultivation globally[2]. This topic is particularly salient in Brazil, where this methodology has been applied. Pilot projects and studies in other countries will be implemented to further refine this methodology to include other global perspectives and references.

In Brazil, the decline in soil carbon stocks due to land use change, deforestation, and/or conventional agricultural practices such as tillage remains a significant concern. These agricultural practices have been widely studied for decades, reflecting their ongoing impact on soil and sustainability in the country[3]. On the other hand, adopting conservation agricultural practices such as no-tillage, crop rotation, rationale irrigation and optimised nitrogen inputs can potentially restore the original levels of soil carbon stocks before anthropic intervention.

Regardless of the soil carbon flux into agricultural systems (i.e. net emissions to the atmosphere or net removals from the atmosphere), some soil carbon-related parameters must be measured over time. Despite the different methodologies currently proposed to measure soil carbon indirectly, such as using statistical modelling and satellite imagery, the direct measurement (i.e., standard) of soil carbon stocks in the fields is the most reliable and widely accepted approach[4]. The rationale for using a widely accepted standard approach is simple and robust: i) soil carbon stock measurements need the quantification of carbon content and bulk density from soil samples that represent a stated layer or depth; ii) soil sampling must be carried out just before the implementation of conservation agriculture practices as a baseline and after a period of time; iii) time period difference results in additionality on soil carbon stocks.

[2] (Clay et al., 2012; Mello et al., 2014; Claassen et al., 2018; Tenelli et al., 2021; de Carvalho et al., 2023; Melo Damian et al., 2023; Vendig et al., 2023)
[3] (Cerri et al., 2007; Mello et al., 2014; Nwaogu & Cherubin, 2024)
[4] (IPCC, 2003)

Given its ease of use for general and direct application on soil carbon stock measurements, this method was identified as the standard approach for the RSB Soil Carbon Impact Indicator Methodology. The purpose of this indicator is to quantify the added soil carbon stocks following the adoption of conservation agricultural practices in sugarcane and grain crops. The RSB Soil Carbon Impact Indicator Methodology may also be applied for monitoring soil carbon stocks over time.

The development of the methodology presented in this report has been conducted following the below project phases:

- Phase 1: Research and literature review (September 2024);
- Phase 2: Soil Carbon Impact Indicator initial draft proposal development (September 2024);
- Phase 3: Online workshop with (inter)national stakeholders and consultations during webinar for RSB stakeholders (October 2024);
- Phase 4: Soil Carbon Impact Indicator final draft proposal development (November 2024).

The above-listed project phases have enabled the study and research of the feasibility and use of a soil carbon impact indicator related to the adoption of conservation agricultural practices in sugarcane and grain crops.

Each project phase has resulted in a report detailing the process followed, the outputs of the project phase, the expert feedback received and any other stakeholder engagements or collaborations held under the framework of the RSB Soil Impacts Collaboration Brazil.

The basis of the methodology used in the soil carbon impact indicator was obtained by the equivalent soil mass method[5], which consists of correcting changes in soil bulk density using baseline equivalent soil mass data as reference[6] . Since soil bulk density increases over time due to no-tilling practices, the bottom section of the topsoil mass needs to be subtracted to ensure equivalent topsoil mass at baseline. This removal section of topsoil mass must then be accounted for in the upper section of the second soil layer, and similarly in the underlying layers[7].

The purpose of developing this methodology was to research and assess the current information, indicators, and tools available (and easily accessible for end users) and build on the existing knowledge and resources to develop a methodology that could serve the needs of RSB, RSB POs, and interested stakeholders.

The RSB Soil Carbon Impact Indicator Methodology for soil carbon gains due to the adoption of conservation agricultural practices in sugarcane and grain crops cultivation has been developed by RSB in collaboration with the third-party consultancy, Imaflora.

[5] (Ellert and Bettany, 1995)
[6] (Fowler et al., 2023)
[7] (Lee et al., 2009)

The indicator methodology presented here was developed through a stakeholder engagement process, where NGOs and national subject matter experts were key stakeholders and contributors.

In line with Principle 8, RSB has developed the RSB Soil Impact Indicator Methodology relating to soil carbon for measuring the effects on soil carbon stocks by adopting conservation agricultural practices as a mechanism to create soil carbon impacts in addition to P&C certification requirements.

In this document the methodology for sugarcane and grain crops feedstocks is presented, however RSB envisions to expand the scope of application of this methodology to also include additional feedstocks in the future.

2. Objective and scope

The objective of developing the RSB Soil Carbon Impact Indicator Methodology is to provide a normative basis for demonstrating additional positive impacts in soil ecosystems – following a science-based and credible approach - through the implementation of appropriate management practices by RSB POs in the RSB system.

This proposed methodology constitutes an additionality to the RSB certification system and would be the basis for POs to make additional sustainability claims beyond standard compliance with RSB traceability requirements. The proposed impact indicator methodology may be utilised by any RSB PO, with a global scope of application.

This proposed methodology will be presented to the RSB Assembly of Delegates (AoD) for approval at their upcoming meeting on the 12 December 2024 to be held in Geneva (Switzerland). Subject to approval from the RSB AoD, RSB Secretariat would then initiate a pilot project in 2025 to test the application of the methodology (and collect data based on which specific certification requirements will be developed and integrated into the RSB certification system).

3. Methodology and implementation for soil carbon stocks

The following information corresponds with the data that is required for the implementation of the methodology.

3.1 Soil sampling in the field

To evaluate soil carbon stocks, the following methodology must be applied for soil sampling in the field. The sampling recommendations for minimum requirement have generalist application in most crops and were based on the technical and scientific recommendations from the Brazilian Agricultural Research Corporation (EMBRAPA - Empresa Brasileira de Pesquisa Agropecuária). However, sampling details covering

positioning, depth and layer in the field have a more specific application and were based on decades of technical and scientific studies for recommendations of soil fertilisation by the Agronomic Institute of Campinas (IAC – Instituto Agronômico de Campinas). For the IAC, soil carbon content is considered a key attribute of soil fertility and is, therefore, consistently analysed across various crops. This approach validates the use of both EMBRAPA and IAC recommendations in the development of the RSB Soil Carbon Impact Indicator Methodology.

a) Minimum soil sampling requirement [*,**,***]:

Type A Project (for smallholders): If the project area is homogeneous and smaller than 75 hectares[8], then at least 3 composite soil samples should be taken.

Type B Project (for large scale-farming): If the project area is greater than 75 hectares, then a stratified random sampling strategy must be applied. Each quantification unit within the project area should be divided into sub-units (i.e., strata) based on factors influencing soil carbon stock distribution (e.g. soil taxonomy, soil parameters, crop history, environmental conditions, landscape[#]) that make each stratum more homogenous than the project area in its entirety. The number of homogeneous sites (i.e. the number of strata) and soil composite samples should be maximised. Therefore, at least 3–5 composite samples should be taken within each stratum[9].

[#] Another variable for stratification may be included as long as there is scientific or technical justification.

[*] Each sampling point must be georeferenced and the soil collected at each point must be considered a simple sample. The simple samples from each stratum must be mixed and homogenised to obtain a unique composite soil sample. The number of simple samples to set one composite sample must follow statistical models that allow a sampling error of less than or equal to 20%. Different statistical models can be applied

[8] RSB small holders definition: https://rsb.org/wp-content/uploads/2024/06/RSB-STD-01-002-v.1.5_RSB-Glossary-of-Terms.pdf

[9] (Verra, 2024)

for soil carbon and soil bulk density because they have discrete spatial variability in the field[10]. ** Applied for all crops. *** This sampling approach differs from the requirements of EU RED II as it was based on EMBRAPA recommendations[11] and considers a more generalist and global approach which can be applied to different cultures and environments.
b) Soil sampling positioning, depth and layers*: b1) Sugarcane Soil sampling positioning in the field should be taken 0.30 m from the center of the sugarcane row in the opposite direction to the fertilization row. This distance represents the spatial variability within the row spacing. Each sampling point shall be taken for the first 0.30 m of soil, and additional measurements might be implemented as desired. b2) Grain and other crops Soil sampling positioning in the field should be taken between the sowing row (i.e. inter-row). Each sampling point should reach at least 0.30 m of soil depth in a single layer, or stratified in layers of 0.10 m (e.g. 0-0.10 m, 0.10-0.20 m and 0.20-0.30 m). The PO can sample deeper layers at 0.10 m in increments. * Soil sampling positioning, depth and layers were based on IPCC[12] guidelines and technical recommendations for soil fertilisation and liming[13].
c) Soil sampling schedule - minimum requirements, can be increased if the PO finds it more viable*,**,***:

[10] (Lawrance et al., 2020)
[11] (Chitolina, 2009)
[12] (IPCC, 2003)
[13] (Cantarella et al., 2022)

c1) Sugarcane**

First sampling (baseline – t): near (just before or after) harvesting the last ratoon in the sugarcane reference cycle (i.e. cycle 1).

Second sampling (t_{+5}): after 5 years of the first sampling

Third sampling (t_{+10}): i) after 5 years of the second sampling

Further sampling: subsequent samplings follow the above model every 5 years.

c2) Grain and other crops

First sampling (baseline – t): before sowing the reference crop.

Second sampling (t_{+5}): after 5 years of the first sampling

Third sampling (t_{+10}): after 5 years of the second sampling

Further sampling: subsequent samplings follow the above model every 5 years.

* Lime must not have been applied in the sampling season due to its composition of inorganic carbon, which can wrongly overestimate the carbon content during soil quantification by the dry combustion method.

** Soil sampling in the cane plant and the first ratoon should be avoided because the soil is recovering its steady state during this period after the disturbances generated in the previous preparation.

*** The soil sampling details and procedures adopted in the baseline must be applied throughout the sampling periods.

d) Equipment/tools and usage of soil sampling *

d1) Disturbed samples

A Dutch auger, probe, or any other coring method can be used as equipment/tool. The disturbed soil samples are used to determine carbon content.

d2) Undisturbed samples

Use a volumetric ring with a minimum 5 cm internal diameter, a minimum 5 cm height, and specific auger as equipment/tools.

The undisturbed soil samples are used to determine soil bulk density. * If a volumetric ring method is impossible, a reliable and scientific-based method should be used instead.

3.2 Assessment in the laboratory

To assess soil carbon stocks, it is necessary to determine the carbon content and the bulk density. The following methodology must be applied in the laboratory:

a) Soil carbon content * a1) Soil treatment Composite samples should be oven-dried at 105 °C for 24 hours, or at 40 °C for 72 hours, and processed in a soil mill to pass through a 2.00 mm sieve and then ground to pass through a 0.15 mm sieve. a2) Soil analysis Composite samples should be determined by a dry combustion method using automated commercial instruments. The results should be expressed as g of C kg^{-1} dried soil[14]. * Alternative techniques for assessing soil carbon content may be accepted, provided they are scientifically robust, internationally recognised, and published in peer-reviewed journals.
b) Soil bulk density *,** b1) Soil treatment Composite samples should be oven-dried at 105 °C for 24 hours.

[14] (Nelson and Sommers, 1996)

b2) Soil analysis Composite samples should be determined by gravimetry (consider the internal volume of the volumetric ring used in the undisturbed sampling). The result should be expressed as kg dm^{-3} dried soil. * If the volumetric ring method is impossible, a reliable and scientific-based method should be used instead. ** Alternative techniques for assessing soil bulk density may be accepted, provided they are scientifically robust, internationally recognised, and published in peer-reviewed journals.

4. Calculation

Soil carbon stocks should be calculated using the following equation:

$$SCS = CC \times SD \times BD \times 10{,}000 \times 0.001$$

Where:

- SCS is the soil C stock (Mg ha^{-1});
- CC is the carbon content in soil (g kg^{-1});
- SD is the soil depth (m);
- BD is the bulk density (kg dm^{-3});
- 10,000 is the conversion factor of m^{-2} to ha^{-1};
- 0.001 is the conversion factor of kg to Mg.

The rationale behind the abovementioned equation used in the soil carbon impact indicator is widely accepted and based on internationally recognised methodologies, such as the IPCC 2003 guidelines.

Further calculations by the equivalent soil mass method[15], which consists of correcting changes in soil bulk density using baseline equivalent soil mass as reference, must be conducted. Since soil bulk density increases over time due to no-tilling practices, the bottom section of the topsoil mass needs to be subtracted to ensure equivalent topsoil

[15] (Ellert and Bettany, 1995)

mass at baseline. This removal section of topsoil mass must then be accounted for in the upper section of the second soil layer, and similarly in the underlying layers[16].

When necessary, soil carbon stocks can be presented as equivalent CO_2 per unit of energy production using the following calculation:

$$SCF = (SCS \times 44/12) / (BY \times 10^6)$$

Where:

- SCF is the soil carbon footprint (g CO_2e MJ^{-1});
- SCS is the soil C stock (Mg ha^{-1});
- 44/12 is the conversion factor from carbon to CO_2;
- BY is biofuel yield (MJ ha^{-1});
- 10^6 is the conversion factor from Mg to g of CO_2e.

To align with the concept of additionality, the soil carbon accumulation (difference in soil carbon stocks) over time should be used for claims. The additionality in soil carbon shall be obtained through the difference in carbon stocks between sampling periods (item 3.1). This difference between initial soil carbon stock at initial sampling and the subsequent stock at t+5 reflects the additional carbon sequestration achieved through sustainable practices.

5. Verification

During verification, the PO must demonstrate documents that prove the performance of soil analyses with the date, laboratory reports and/or technical reports on the results. The results must be verified by checking that the pertinent documentation, processes and systems are in place (such as for example electronic spreadsheets including the required data).

Table 1 below includes a list of recommended verification methods to be used in the auditing of the implementation of this methodology.

Table 1: Verification methods

[16] (Lee et al., 2009)

	Item	Verification Method
Soil sampling	Minimum soil sampling requirement *, **, ***, ****:	Verifiable through records, photos, spreadsheets and other documents if the samples were collected homogeneously and in accordance with the requirements of the criteria.
	Soil sampling positioning, depth and layers*:	Verifiable through records, photos, spreadsheets and other documents demonstrating that the collection was made considering the recommended positioning, depth and layers.
	Soil sampling schedule*: To be continued sampling:	Verifiable through records, photos, spreadsheets and other documents demonstrating that the collection was made according to the recommendation.
	Equipment/tools and usage of soil sampling	Verifiable through records, photos, spreadsheets and other documents demonstrating which equipment was used for the collections.
Laboratory	Soil carbon content	Verifiable through a laboratory report demonstrating the methods used.
	Soil bulk density	Verifiable through a laboratory report demonstrating the methods used.
Calculation	Equation	Verifiable through the verification of the calculation of the equation
Capacity	Skilled staff of PO	Verification through training records, interviews, and internal audit documentation.

Potential implementation challenges:

The implementation of the RSB Soil Impact Indicator Methodology presents some potential challenges, such as:

- Methodology differences per type of feedstock: Certain feedstocks, such as for example sugarcane crops, require many specifications not considered in the methodologies for other crops/feedstocks;
- Data reliability vs. costs: The higher the reliability of the data on soil carbon (and the various parameters to be analysed), the higher the costs involved;
- Complex sampling periods: The tillage of sugarcane soil complicates the scheduling of sampling periods. Other crops may also have complex growth cycles or rotational patterns that affect the timing when the sampling can be conducted.

- Verification sampling methods: positioning in soil sampling can be relevant across various row crops, e.g. the soil sampling positioning for sugarcane (along 0.30 m from the centre of the row) could lead to subjective data collection.

6. Conclusions

The RSB Soil Carbon Impact Indicator related to the adoption of conservation agricultural practices has been developed primarily with the local partners of the RSB Soil Carbon Impacts Collaboration in Brazil: Nuseed and Raizen as RSB members and industry partners, and Imaflora as technical consultants and for stakeholder engagement.

RSB has developed a methodology to calculate soil carbon gains by adopting conservation agricultural practices (Table 2). This approach aims to incentivise RSB-certified POs to generate positive environmental soil impacts (promoting the enhancement of carbon content in soil) in addition to the sustainability impacts driven through the implementation of RSB certification requirements across supply chains.

Table 2: Non-exhaustive list of conservation agricultural practices and the derived expected positive impacts on soil carbon stocks.

Conservation agriculture practices	Description	Expected impact on soil carbon stocks
No-tillage or reduced tillage	growing crops without disturbing the soil through tillage	• soil carbon content: increase • soil bulk density: decrease
Crop rotation	planting different crops sequentially on the same field seeking the improvement of soil attributes and agrosystem resilience, optimize nutrients in the soil, and combating pest and weed pressure	• soil carbon content: increase • soil bulk density: decrease
Rationale irrigation	water supply through methods appropriate to each crop and guided by monitoring the plant demand.	• soil carbon content: increase
Optimized nitrogen fertilization	nitrogen application is done through methods	• soil carbon content: increase

	appropriate to each crop and guided by soil supply and plant demand.	
Organic amendments	partial or total replacement of inorganic fertilizers by organic ones.	• soil carbon content: increase

By doing so RSB provides an innovative opportunity to link market participants by insetting sustainability impacts and responding to both a growing need for reporting on sustainability footprints in global supply chains, as well as ongoing demand for support of locally applied measures to counter the negative impacts of conventional (resource intensive) agricultural practices.

For the development of this methodology, a stakeholder workshop with national and international participants was held in October 2024 to ensure a transparent and inclusive stakeholder engagement process.

The sampling recommendations were based on the technical and scientific recommendations from the Brazilian Agricultural Research Corporation (EMBRAPA - Empresa Brasileira de Pesquisa Agropecuária and the Agronomic Institute of Campinas (IAC – Instituto Agronômico de CampinasTo further enhance the RSB methodology's applicability, a review and comparison analysis with other methodologies was conducted (such as EU RED and Verra), with the aim to identify differences and streamline the requirements set under the framework of this methodology. The results of the comparison analysis with other methodologies can be found in Table 3 below.

Table 3: Differences between soil sampling methodologies

Soil sampling detail or procedure	RSB Soil Impact Indicator Methodology	EU RED (2022/996) [17]	Verra (VM 0042) [18]
Minimum requirement	If the project is smaller than 75 hectares, at least 3 composite soil samples should be taken. If the project area is greater than 75 hectares, the stratified random	At least one composite sample of 15 well distributed simple samples per every 5 hectares or per field, whichever is smaller	Stratified random sampling strategy with 3-5 composite samples within each stratum

[17] (EU Commission, 2022)
[18] (Verra, 2024)

	sampling strategy must be applied.		
Sampling period	5 years at minimum or more frequently	5 years at minimum or more frequently	5 years at minimum or more frequently
Soil depth	0.30 m at minimum	0.30 m	0.30 m at minimum
Crops	Sugarcane and other grain crops	no restriction	Cropland or grassland
Soil carbon analysis	Dry combustion method	Dry combustion method	Several
Soil bulk density analysis	Preferably volumetric ring	Preferably volumetric ring	Preferably volumetric ring

As with the introduction of any innovation within a globally recognised leading sustainability certification system, the implementation within the RSB certification system and the full integration into all relevant system elements, including the RSB Book & Claim Impacts registry, will be subject to continuous learning and improvement.

The aim of releasing this first version of the RSB Soil Carbon Impact methodology is to have a sufficiently robust impact indicator to be further developed over time, reflecting the needs of all parties involved in RSB certification. RSB Secretariat believes that by having this methodology in place, the RSB system will be providing a science-based approach to demonstrating positive impacts on soil systems in line with the RSB mission of promoting the transition to a biobased economy.

7. References

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8. Abbreviations

BD – Soil bulk density

BY – Biofuel yield

CC – Soil carbon content

CO_2 – carbon dioxide

EMBRAPA - Empresa Brasileira de Pesquisa Agropecuária

IAC – Instituto Agronômico de Campinas

IPCC – Intergovernmental Panel on Climate Change

NGOs – Non-governmental organizations

P&C – Principles and criteria

PO – Participating operators

RED II – Second Renewable Energy Directive

RSB – Roundtable on Sustainable Biomaterials

SCF – Soil carbon footprint

SCS – Soil carbon stocks

SD – Soil depth

9. Terms and definitions

Conservation agriculture practices: sustainable practices that involve a combination of techniques to improve soil health, increase crop yields, and reduce environmental impact, such as no-tillage, crop rotation, rationale irrigation, optimised nitrogen fertilization, and organic amendments.

Disturbed soil samples: soil sampling is carried out using an auger that disaggregates the soil bulk mass.

Equivalent soil mass: for management-induced changes in soil carbon storage to be assessed reliably, the masses of soil being compared must be equivalent.

Grain crops: the crops that produce grains regardless of harvesting (e.g., soybean, maize, rice, barley, wheat, carinata, rapeseed, crotalaria, pigeon pea, etc.).

Homogeneous field: field that contain the same soil taxonomy, crop history, environmental conditions, landscape.

Large-scale farming: feedstock producers (e.g. farmers) whose total amount of land does exceed 75 hectares.

Project area: an area that may consist of multiple farms, provided it is homogeneous in terms of soil characteristics.

Sampling season: the current year of soil sampling (e.g., first sampling, second sampling, third sampling and so on).

Smallholders: feedstock producers (e.g. farmers) whose total amount of land does not exceed 75 hectares. Note: The RSB will continue to explore possible options for the definition of smallholders, based on additional parameters such as the use of family labor, access to technology, income, etc.

Soil bulk density: the amount of soil mass per unit of soil volume.

Soil carbon content: the amount of carbon per unit of soil mass.

Soil carbon stocks: the amount of carbon per unit of area considering certain soil layer.

Stratum: a subset of each quantification unit within which the value of a variable, and the processes leading to change in that variable, are relatively homogenous.

Sugarcane cycle: the sugarcane period in which several harvests occurs, generally 5-7 years.

Sugarcane ratoon: the sugarcane seasons after the first harvest.

Undisturbed soil samples: soil sampling is carried out using volumetric rings into specific auger such that the soil aggregation and integrity is preserved resulting in a core of soil bulk mass.